

Yuri Shiroiva Berbereia

AI Engineer / Data Scientist

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PROFESSIONAL SUMMARY

AI Engineer and Data Scientist with 3+ years of experience building generative AI systems and data pipelines in production, with a focus on data quality, traceability, and governance. Technologist Degree in Applied Artificial Intelligence from PUCPR, with honorable mention.

TECHNICAL SKILLS

Generative AI & LLMs: production LLM applications, multi-agent systems, Claude on Vertex AI, MCP (Model Context Protocol), prompt engineering, structured output with JSON Schema, RAG, LangChain, Hugging Face Transformers

Machine Learning: PyTorch, scikit-learn, gradient boosting, CNNs, semantic segmentation (U-Net, U-Net++, DeepLabV3+), ensembles, multi-task learning, cross-validation, bootstrap confidence intervals

Data: Python, SQL, pandas, NumPy, ETL pipelines, data quality, Apache Cassandra, PostgreSQL, bi-temporal data modeling, time series, geospatial data (GeoJSON, Sentinel-2)

Cloud & Tools: GCP (Vertex AI, IAM, VPC), Docker, Git, REST APIs, Streamlit

Domain: agricultural machinery telemetry, CAN bus / ISOBUS (J1939, ISO 11783, ISOXML), precision agriculture

PROFESSIONAL EXPERIENCE

Data Scientist / AI Engineer — IoTag (IoT platform for AgTech)

2024 – Present

Responsible for the AI layer and data pipelines of an agricultural telemetry platform that ingests 5,000+ raw CAN bus files per day from 500+ machines in operation.

- Cut signal decoding for new machine models from 1 month to 2 weeks by architecting a multi-agent LLM system with tool calling via MCP that validates each signal against ground truth and hold-out data before deployment; 20+ signals now in production.
- Reduced machine fault diagnosis time from 2 weeks to 5 days by designing and shipping a generative AI feature (Claude on Vertex AI) that analyzes operation telemetry and delivers diagnostics as structured output validated by JSON Schema.
- Brought traceability and auditability to the configuration of every field operation by modeling a versioned history in Cassandra (bi-temporal, append-only) that reconstructs the state of any operation at any point in time.
- Fixed distortions in applied-product totals reported to customers by diagnosing two defects in the production ingestion pipeline (unit conversion and aggregation scope).
- Standardized data from 10+ machine brands (J1939/ISOBUS) into a modular Python ETL pipeline that underpins the platform's analytics and AI features.
- Automated the generation of ISOXML files and GeoJSON coverage maps from raw telemetry, replacing manual assembly for each operation.
- Integrated into the pipeline the geospatial data of 200+ fields that were locked in a vendor's proprietary format, with no dependency on the partner's API.

Software Development Intern — Optcore

04/2023 – 05/2024

- Built Python data quality routines (cleaning, standardization, and deduplication of heterogeneous sources) for a retail analytics project and evolved the Streamlit and Appsmith dashboards that a client's leadership used to track about 30 accounts.

PROJECT

Field Boundary Segmentation with Deep Learning (Undergraduate Thesis) — PUCPR

2026

- Outperformed the semantic baseline of the FTW-Brazil benchmark in Panoptic Quality (0.51 vs. 0.46) for field segmentation in Sentinel-2 imagery, using a weighted ensemble of 6 networks (U-Net, U-Net++, and DeepLabV3+ with ResNet-34/50 encoders) and TTA, reaching a mean Dice of 0.77 (median 0.87).
- Deployed the solution at IoTag as a university extension project, where 5 professionals from engineering, data science, and DevOps rated it 4.6/5 on average for usefulness and accuracy.
- Reached a mean precision of 0.79 by calibrating the decision threshold and removing spurious components in post-processing, and validated the results with 95% confidence intervals and effect size (Cohen's d).

EDUCATION

- **Technologist Degree in Applied Artificial Intelligence** — PUCPR | 2024 – 2026. GPA 9.27/10. Honorable mention for overall academic performance.
- **Electrical Engineering (Robotics and Industrial Automation)** — UFJF | 2021 – 2023. Not completed; transferred to PUCPR in 2024.

CERTIFICATIONS & LANGUAGES

Certifications: IBM Generative AI Engineering (2026) · IBM Generative AI Engineering with LLMs (2026) · Supervised Machine Learning, DeepLearning.AI (2025) · IBM Data Science Professional (2024) · Google Data Analytics (2023)

Languages: Portuguese (native) · English (proficient technical reading, basic conversation) · Spanish (basic)